

s073e — Deep audit of Robinson & Frueh 2025
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Part 1: implementation lessons we can adopt.
Part 2: does their work encroach on our s073 framing?

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Abstract

Robinson & Frueh 2025 (RF25 hereafter) is the highest-remaining-threat paper flagged by the s073c literature-novelty audit: a 37-page treatment of global light-curve attitude inversion under measurement noise and inertia uncertainty, by the closest competitor group (Frueh, Purdue). This document is a deep read of RF25 with two narrow goals. **Part 1** catalogues every implementation element of RF25 that could materially improve our pipeline, with concrete decisions on whether to adopt, adapt, or pass. **Part 2** asks whether their work encroaches on the specific framing s073 is hypothesising — that the body-frame Casimirs ($|\mathbf{L}|, 2T$) plus polhode phase parameterise an inertial- $\mathbf{L}_{J2000}$ ambiguity that is generic to asymmetric-inertia bodies. The conclusion of Part 2: the framing space is *not* closed by RF25, but they have observed adjacent phenomena (continuous ω -bands on axisymmetric bodies) that we must explicitly distinguish from our cat-4 claim if we want to publish.

Reading guide

Treat this as a working document, not a paper. It is more thorough than a reading note because we may end up citing or reproducing some of these claims; it is less polished than a manuscript section because the cat-4 framing is still a hypothesis (s073d is ambiguous as of commit b6e3482). Numbers and quotes come from the PDF at `/home/girish/Documents/s40295-025-00557-9.pdf`; specific page citations use the *print* page numbers shown at the top of each PDF page (e.g. “p. 5627” = the journal’s page 5627, not the PDF’s nth sheet). Citations to RF25’s own bibliography use their numbering ([RF#nn]).

1 What RF25 is, in one paragraph, before we judge it

RF25 builds a global light-curve attitude-inversion pipeline that explicitly optimises through (a) realistic per-timestep measurement noise and (b) uncertainty in the object’s principal moments of inertia. The state vector is $\mathbf{x} = (p_1, p_2, p_3, \omega_1, \omega_2, \omega_3, J_y/J_x, J_z/J_x)^T$ — three MRP (modified Rodriguez parameter) attitude components, three body-frame angular velocity components, and two inertia-tensor ratios. The loss is a negative-log-likelihood under a per-timestep Gaussian model. The solver is parallel BFGS started from 10^5 uniform initial guesses (uniform on $SO(3)$, uniform in an ω -ball, Gaussian on inertia ratios). The forward model uses a Blinn-Phong BRDF and a semi-analytic Sutherland-Hodgman polygon-clipping shadow algorithm precomputed into a 4D body-frame “shadow cache.” They demonstrate on two objects: an ATLAS V rocket body (1987-084D, axisymmetric inertia) and a HYLAS-4 box-wing satellite (2014-004A, asymmetric inertia). They run with both low ($\sigma_J = 0.1$) and high ($\sigma_J = 1.0$) inertia-ratio prior uncertainty. All cases use a model mismatch between the truth shape and the assumed inversion shape, deliberately introduced to simulate real-data conditions.

Part I

Implementation lessons we can adopt

Twelve elements of RF25 stand out as candidates for transplantation into our pipeline. We sort them by “how much they would change our code” and tag each with a **decision** of ADOPT (drop in directly), ADAPT (useful idea, our regime differs), EVALUATE (worth a small experiment to test), or PASS (their constraint, not ours).

1.1 The negative-log-likelihood (NLL) loss with per-timestep σ_k weighting and signal-rescaling [ADOPT, high-value]

RF25’s loss function is their single most exportable contribution to our work. Their derivation (their §2.1, p. 5–6 of the article):

$$f_{\text{NLL}}(\hat{\mathbf{S}}, \mathbf{S}) = \frac{1}{m} \sum_{k=1}^m \left[\ln \sigma_k + \frac{1}{2} \left(\frac{S_k - \hat{S}_k}{\sigma_k} \right)^2 \right] \quad (1)$$

where $\hat{S}_k$ is the predicted ADU flux at timestep k , S_k is the measured ADU flux, and σ_k is the predicted measurement standard deviation (computed from the full noise model in their §3.5.1; see Item ??). The $\ln \sigma_k$ term penalises solutions that “predict noisy regions confidently”; the squared term is the usual chi-square.

Why we should care. Our current cost surfaces use surrogate MSE on magnitude. RF25 argue (their p. 4–5) that magnitude-space ℓ_2 “attributes much larger errors to brighter regions of the light curve, leading to solutions that overfit bright specular glints while ignoring smaller variations in the signal,” and is blind to per-timestep noise. *This is precisely the bright-peak overfit pattern we have seen* — e.g. s058 / s059 LM polish on a 21-epoch window converging to phantom basins disagreeing with the full-LC hi-fi by $\rho = 30$ –70 (see `memory project.local.window.polish.phantom.basins.md`).

A subtle but load-bearing trick they add is the **signal rescaling**: before computing the loss, $\hat{\mathbf{S}}$ is multiplied by $\|\mathbf{S}\|/\|\hat{\mathbf{S}}\|$, forcing the prediction and observation to have equal ℓ_2 norm in flux (their p. 5). This absorbs systematic albedo-area errors and prevents the loss from penalising LCs of the correct shape but wrong scale. *This is essentially a free correction for the kind of BRDF coefficient uncertainty that haunts our m048 forward model.*

Concrete adoption path. Replace the surrogate-MSE residual in `lib/forward.py` (and in the LM cost in s058/s059 polish stages) with f_{NLL} on ADU flux, computed with our existing per-timestep σ_k (`lib.noise` already returns this for the m048 noise model). Add the $\|\mathbf{S}\|/\|\hat{\mathbf{S}}\|$ rescaling factor inside the residual. This is a ~ 30 -line change. No retraining of the surrogate is needed because the surrogate maps body-frame coordinates $\rightarrow$ magnitude, not loss-space; we just re-score on top of the same evaluator.

Caveat. Our ρ -band convention is rooted in surrogate MSE ratios, not NLL. If we adopt NLL we either (a) report both during a transition, or (b) re-derive the band thresholds. Either is cheap.

1.2 The full sensor + sky noise model (Eqs. 35–40) [EVALUATE, scoped to real-data work]

RF25 enumerate *four independent* noise contributors and combine them per timestep (their §3.5.1, p. 17–19):

- **Background Poisson noise** (their Eq. 37): $\lambda_{\text{back},k} = n_{\text{pix}} \cdot (\lambda_{\text{moon}} + \lambda_{\text{star}} + \lambda_{\text{twi}} + \lambda_{\text{zod}} + \lambda_{\text{air}} + \lambda_{\text{poll}})$. This is scattered moonlight, integrated background starlight, twilight, zodiacal light, airglow, and light pollution, each as an independent Poisson term.

- **Scintillation noise** via Young’s atmospheric approximation (their Eq. 38): a function of telescope aperture diameter D , exposure time Δt_k , observation zenith angle γ_k , and observing station altitude h_{obs} .
- **Sensor noise** (their Eq. 35): dark current (Poisson, rate λ_{dark} per pixel per second) plus read noise (Gaussian, σ_{read}). The flat-field term scales as the square root of total signal across pixels (their Eq. 36).
- **Shot noise** on the object signal (their Eq. 39): $\lambda_{\text{shot},k} = \bar{S}_{\text{SO},k}/\sqrt{g}$.

Total per-timestep variance is their Eq. (40): $\sigma_{S,k}^2 = \lambda_{\text{back},k} + \bar{S}_{\text{SO},k}\sigma_{Y,k}^2 + \lambda_{\text{shot},k} + \sigma_{\text{flat}}^2 + \sigma_{\text{sensor}}$.

Why we should care. Our current noise model uses a fixed apparent-magnitude noise floor (the standard $m_{\text{noise}} = 0.05$ mag). RF25 demonstrate (their Fig. 6c, 12c) that the noise envelope varies by 4–5 $\times$ across a single light curve: faint epochs are scintillation-dominated, bright epochs are read-noise dominated. *This is precisely the regime where the f_{NLL} loss buys you something over surrogate MSE.*

Decision. EVALUATE not ADOPT, because our entire benchmark cohort (m048 seeds 0–119) was generated with the simpler noise model. Swapping to the full noise model mid-cohort would invalidate the post-fix replications (s068–s071) we just committed. The right path is:

1. Implement the RF25 noise model in a new module `lib/noise_rf.py` that returns σ_k as a function of (geometry, telescope params, flux level). Wire it as an optional flag.
2. Generate a *small* replication cohort (10 seeds, RF-noise) and verify the surrogate still resolves Band A–C basins. The surrogate is bridge-independent (memory `feedback_surrogate_is_bridge_i` so it should not need retraining.
3. If basin structure changes materially, the full noise model becomes a secondary publication thread; if not, fold it in for the real-data chapter only.

1.3 The shadow cache [EVALUATE, high-leverage if forward-model wall is the bottleneck]

RF25 build a 4D boolean lookup table of shape $n_{\text{Az}} \times n_{\text{El}} \times n_f \times n_f$ indexed by (azimuth, elevation, shadower face, shadowed face) in the *body frame*. Their RF Eq. (??):

$$\hat{\mathbf{e}} = \begin{bmatrix} \cos(\text{El}) \cos(\text{Az}) \\ \cos(\text{El}) \sin(\text{Az}) \\ \sin(\text{El}) \end{bmatrix} \quad (2)$$

At evaluation time, the body-frame Sun and observer directions are converted to (Az, El), rounded down (their floor-operator scheme, Eqs. 28–29) to the nearest cache entry, and the per-face shadowing is read directly. At $n_{\text{Az}} = n_{\text{El}} = 200$ and $n_f = 100$, the cache is **50 MB** (one bit per entry). They report the truncation error to nearest cached cell is “negligible” because the only faces that flip due to the rounding are those that are nearly tangent to the illuminator.

Why we should care. The shadow computation is currently a notable fraction of our forward-model wall time. For Path 2 (s072 closed-form $q(t)$) in particular, where the dynamics step is now microseconds, shadow ray-casting becomes the dominant cost. The s073b cprofile shows the ϕ -ODE is 81% of Path 2 wall (`experiments/s073b_path2_cprofile.md`); after we replace that with `elliprj`, shadow cost rises in the ranking.

Decision. EVALUATE. Build a one-object prototype on m048 geometry, measure speedup vs the trimesh batched ray cast. If the speedup is $\geq 5\times$ on a representative LC, integrate; if less, the engineering cost (cache invalidation when components articulate) is not worth it.

Caveat for articulated geometry. m048 has solar panels that sun-track (`src/articulation/behaviors/`). RF25’s cache assumes *static* geometry in the body frame. We would need either (a) separate caches per panel orientation, or (b) compute static-component shadows from the cache and articulated-component shadows from ray casting. Option (b) matches how our current shadow

engine already factors the mesh (cf. `src/computation/shadow_engine.py`’s static/articulated separation), so adoption is mechanical.

1.4 Semi-analytic Sutherland–Hodgman shadowing [PASS for now]

The clipping algorithm RF25 use to compute the shadowed area on each face (their §3.3.1–3.3.3, p. 11–14) replaces pixel-wise ray casting with analytic polygon clipping. Eq. 24 is the inclusion–exclusion formula for mutual higher-order shadow intersections:

$$\bar{a} = a - \sum_{d=1}^n \sum_{K \in \text{comb}(n,d)} (-1)^{d+1} A\left(\bigcap_{k \in K} \mathcal{P}_k\right) \quad (3)$$

truncated at depth $d = 2$ in their work. They report “nearly identical” light curves to pixel-wise reference (their Fig. 6d and 12d).

Decision. PASS. We already use trimesh’s batched ray intersection, which is well-optimised and handles arbitrary non-convex geometry without the truncation–depth trade-off. The Sutherland–Hodgman approach is most useful when the geometry is exactly polygonal (which RF25’s simplified models are); our m048 STL is a 30k+ triangle mesh where the inclusion–exclusion combinatorics grow uncomfortably. The shadow-cache (Item ??) likely subsumes the speedup that would motivate switching the underlying algorithm.

1.5 State parameterisation: MRP attitude + inertia ratios [EVALUATE for asymmetric–inertia uncertainty work; PASS for our main pipeline]

RF25 search over $(p_1, p_2, p_3, \omega_1, \omega_2, \omega_3, J_y/J_x, J_z/J_x)$, with $J_x \equiv 1$. Two pieces here.

(a) MRP vs quaternion. The MRP RF Eq. (??) maps to the DCM via $C(\mathbf{p}) = I_3 + (8\tilde{p}\tilde{p} - 4(1 - \|\mathbf{p}\|^2)\tilde{p})/(1 + \|\mathbf{p}\|^2)^2$ (their Eq. 9). MRPs avoid the unit-norm constraint of quaternions and are 3-dimensional, simplifying BFGS. They have a singularity at the antipode of identity which RF25 handle via shadow MRPs (their reference [37]).

Decision: pass. Our quaternion infrastructure is mature, the propagator-fix work hardened it, and the convention bug experience makes us cautious about changing representations. The MRP advantage is mainly for unconstrained optimisation, which is not our bottleneck.

(b) Inertia ratios in the search state. This *is* interesting. RF25 directly optimise J_y/J_x and J_z/J_x jointly with (q, ω) . For real-data work on objects without precise inertia knowledge, this is the right architecture. Their Case 2a (box-wing, low uncertainty) achieves 0.78% median $|\omega(0)|$ error; Case 2b (high uncertainty) achieves 2.36%. *The inertia uncertainty does not prevent inversion convergence on the asymmetric body.*

Decision: evaluate. For the m048 cohort (known inertia by construction) this is unnecessary. But the Frueh group’s framing is “real-data inversion needs inertia-ratio search” and ours is implicitly “we know the inertia exactly.” If we want a real-data chapter we will need this. Slot it for after the cat-4 framing is settled.

1.6 Loss-based convergence classification: the α threshold [ADAPT]

RF25 define f_{best} as the loss obtained by placing the truth state in the assumed-model forward simulation (their §2.5.1, p. 8). Then: solutions with $f_i < f_{\text{best}}$ are flagged as *overfits* (discarded); solutions with $f_{\text{best}} < f_i < \alpha \cdot f_{\text{best}}$ are accepted, with $\alpha \in [1.5, 2]$ chosen by the user.

Why we should care. This is structurally the same construct as our ρ -band convention but with two important differences:

- It explicitly *filters out overfits* (any solution below the truth-model loss). We do not do this. We have spent significant debugging time on solutions that scored “better than truth” because of surrogate-vs-hi-fi gap or noise overfitting.

- It is parameterised by an interpretable α that the user can loosen if the model mismatch is high (RF25 used $\alpha = 2$ for the matched-shape cases and $\alpha = 1.5$ for the box-wing cases with heavier mismatch).

Decision: adapt. Define an analogous threshold in ρ -space: $\rho_{\text{best}} := \rho(\hat{S}(\text{truth}), S_{\text{obs}})$, discard $\rho_i < \rho_{\text{best}}$ as overfits, accept up to $\rho_i < \alpha \cdot \rho_{\text{best}}$ for some α calibrated to the m048 noise level. This would have caught the s058 oracle-injection artefact earlier.

1.7 Initial-condition sampling: uniform $SO(3)$ + $r^{1/3}$ -scaled ball for ω [EVALUATE]

RF25’s sampling (their §2.3, p. 6):

- **Attitude:** sample $(g_1, g_2, g_3, g_4) \sim \mathcal{N}(0, 1)$ independently, normalise to unit length; this gives a uniform quaternion on S^3 (Marsaglia 1972). Then convert to MRP via RF Eq. (??) $\mathbf{p} = \mathbf{q}_{1:3}/(1 + q_4)$. The uniform- $SO(3)$ property holds for our quaternion grid too if we sample S^3 uniformly.
- **Angular velocity:** sample $r \sim \mathcal{U}(0, 1)$, scale by $r^{1/3}$, multiply by ω_{max} , and direct uniformly on S^2 (again Marsaglia). The $r^{1/3}$ factor gives uniform volume density inside the 3-ball (their Eq. 8).

Why we should care. Our current ω sampling is a Fibonacci direction grid $\times$ magnitude grid (`project_omega_grid_architecture.md`). This is structured, deterministic, and aligned with the polhode geometry, but *biases magnitude* (uniform over the magnitude grid is not uniform over the ball). RF25’s uniform-ball sampling is unbiased.

Decision: evaluate. For surrogate-scoring (where the cost is per-sample) Sobol or Fibonacci with explicit magnitude weighting is fine. For BFGS or LM polish starts (where the cost is per-converged-point), uniform-ball is more honest. Add as an option, A/B-test on a 10-seed cohort slice.

1.8 The BFGS solver with 10^5 parallel starts in Taichi [PASS for now, EVALUATE for GPU port]

RF25 use BFGS from 10^5 initial conditions, compiled in Taichi for parallel execution (their §2.4–2.5, p. 7). The full Case 1a took 9.8 minutes on an Apple M1 CPU. Comparable: our s059k with $N_{\text{DIRS}} = 800$ Fibonacci grid $\times$ 50 polish starts is 105–127 min per seed on Pool(24+8) (memory `project_s059k_cohort_viable_band_A.md`). The factor- ~ 10 speedup they report is real, but most of it is from (a) the simpler forward model and (b) Taichi’s GPU-compiled inner loop.

Decision. PASS on BFGS itself — our LM polish is well-tuned and the BFGS Hessian approximation is not better in this regime. EVALUATE the Taichi port if the m048 forward model becomes the rate-limiter again.

1.9 Inertia ratios visualisation [ADOPT as analysis tool]

RF25 plot the converged $(J_y/J_x, J_z/J_x)$ histogram (their Fig. 8b, 10b, 13b, 14b). For the axisymmetric rocket body the inertia-ratio cloud spreads along the J_z/J_x axis (Case 1a, Fig. 8b); for the asymmetric box-wing it collapses onto truth (Case 2a, Fig. 13b). This is a clean diagnostic for “does the LC discriminate the inertia tensor.”

Decision: adopt as a plot we should add to the s073d / s074 diagnostic kit. If our hypothesised cat-4 family at fixed $(|\mathbf{L}|, 2T)$ exists, it should appear as a particular curve in the (J_y, J_z) -ratio plane (because $|\mathbf{L}|^2 = \sum_i J_i^2 \omega_i^2$ and $2T = \sum_i J_i \omega_i^2$ couple them).

1.10 Solution-space visualisation in body-frame $(\omega_x, \omega_y, \omega_z)$ [ADOPT]

RF25’s Fig. 7b is a 3D scatter of converged ω solutions in body frame, with the truth in red. The cylindrical bands they describe (axisymmetric case) are visually obvious in this plot. For the asymmetric case (Fig. 8d, 13d, 14d) they plot $(\omega(0)$ azimuth, $\omega(0)$ elevation) as a 2D density histogram.

Decision: adopt. We have produced similar plots ad hoc in s053 and s059i; standardise the plot script so every multi-solution analysis ends with this view. For cat-4 specifically, the polhode signature (closed curve in body frame ω) is the right thing to look at, and this is the canonical view.

1.11 The “averaging destroys solution-space structure” point [ADOPT as policy]

RF25’s §4.3, p. 32: “While it may be attractive to apply an averaging procedure for the initial orientations or angular velocities to produce a single state vector, doing so will generally destroy the structure of the solution space spanned by the original solution set. Instead, we recommend that densely clustered regions of the solution space may be thinned without impacting this structure.”

Decision: adopt as policy. This crystallises a discipline we have been intuitively following but never written down. Add to memory under `feedback_multi_solution.md`.

1.12 Ground-truth f_{best} requires placing truth in the assumed model [ADOPT as gate]

Subtle but important. RF25 do not compute f_{best} from truth in the truth-model; they compute it from truth in the *assumed* (inversion) model. The reason: when there is model mismatch, the truth state in the truth model produces a noiseless LC that, when scored against noisy observations, yields a residual purely from noise. But the truth state in the assumed (mismatched) model yields a residual that captures both noise *and* model-mismatch bias — which is the right denominator for “how good can any solution legally be.”

Decision: adopt. Our “noise floor” for ρ is set by the m048 noise model alone; we should add an analogous quantity that places truth in a deliberately-perturbed forward model (e.g. surrogate v2 vs hi-fi) and use that as the discard threshold for overfits.

Part II

Does RF25 encroach on the s073 framing?

This is the question s073c flagged as load-bearing. RF25 is the closest competitor paper to the cat-4 framing; if it pre-empts us we redirect.

1.13 The s073 framing, stated precisely so it can be compared

What s073 measured (verified, source `notebooks/inversion/survey/results/s073/summary.json`): on m048 seed 89, the Band A polish output `cluster_457` and the truth attitude have $|\mathbf{L}|$ matching to 0.55% but $\mathbf{L}_{J2000}$ -directions differing by 128.6° at $t = 0$, with hi-fi $\rho = 0.904$ (within the noise floor). The s073 framing extrapolates this single observation into a structural claim:

Hypothesised cat-4 ambiguity: *For a torque-free rigid body with no shape symmetry, an LC computed under a fixed inertial geometry $(\hat{u}, \hat{s})$ over a short observing arc depends on the body-frame sun/observer trajectory only — which is determined by $(|\mathbf{L}|, 2T)$ plus a polhode-phase scalar, not by the inertial orientation of $\mathbf{L}_{J2000}$. Thus there exists a continuous 2-parameter family of inertial $\mathbf{L}_{J2000}$ -directions producing similar LCs.*

The framing has two distinct claims, which we must judge separately against RF25:

Claim A (mechanism) LC depends on body-frame geometry only; therefore the inertial direction of $\mathbf{L}_{J2000}$ is (to leading order) unobservable on a short arc.

Claim B (cardinality) The ambiguity is *continuous* (a 2-parameter manifold over the S^2 of $\mathbf{L}_{J2000}$ -directions, modulo discrete flip).

Claim C (generality) This applies on *asymmetric-inertia* bodies — it is not the cat-2 “visual symmetry” phenomenon.

We assess each below.

1.14 What RF25 actually report about solution-space geometry

RF25 enumerate ambiguities through their empirical solution clouds, not through theoretical analysis. The relevant passages and figures:

(p. 23, Case 1a, axisymmetric rocket body, $\sigma_J = 0.1$.) Quote: “Figure 7a shows that the initial state MRP solutions lie on **two disjoint tubes**. The two families of orientation solutions correspond to reflecting the assumed cylinder about the phase angle bisector plane spanned by the vector halfway between the observer and Sun and their cross product — **identical to the mediatrix plane identified in [33]**.”

This is the cat-1 flip problem (Burton–Robinson–Frueh 2024). The two “tubes” are the two sides of the flip mirror.

(p. 23–25, Case 1a continued.) Quote: “Figure 7b reveals that the angular velocity solutions lie mainly along **four cylindrical sections symmetric about the body’s axis of symmetry to maintain a constant precession rate**.”

The cylinders are continuous 1D bands in body-frame ω . RF25 attribute them to the axisymmetric-inertia structure of this rocket body: rotation about the symmetry axis is dynamically free (the precession rate is set by $|\omega(0)|J_t/J_a$ from their Eq. 14, while the spin rate $\omega_s = (J_t - J_a)/J_t \cdot \omega_a(0)$ contributes nothing to the LC when the symmetry axis is also a geometric axis of symmetry). *This is the cat-2 “visual symmetry” phenomenon, in its axisymmetric form.*

(p. 25, Case 1a, spin rate observability.) Quote: “The spin rate of this axisymmetric object about its axis of symmetry is weakly observable in the light curve (at best), we cannot reliably determine the angular velocity magnitude or the axial moment of inertia. This observed ambiguity does not contradict well-accepted methods for angular velocity determination that use brightness ratios [77] or the periodogram [78], which do not attempt to estimate axial rotation rates and assume the object’s attitude is well-approximated as a single-axis spin.”

Median spin-rate error: -95.60% (Table 7) — basically undetermined.

(p. 27, Case 2a, asymmetric box-wing.) Quote: “Compared to Case 1, there is better convergence in the inertia tensor ratios as the object has three unique moments of inertia and no geometric axis of symmetry, **removing the spin rate ambiguity seen in the rocket body cases**. While this light curve does not uniquely determine the initial angular velocity direction, the angular velocity magnitude estimates have a median close to the truth.”

Median $|\omega(0)|$ error: 0.78% (Case 2a, Table 10), 2.36% (Case 2b, Table 11).

(Figs. 13c–d, 14c–d.) For the asymmetric box-wing, the initial orientation-angle error scatters across 0° – 180° with no clear clustering structure visible at the resolution of their 2D histograms. The $(\omega(0)$ azimuth, $\omega(0)$ elevation) plots also show scattered solutions across both hemispheres without obvious continuous bands.

(p. 33, Conclusion, summary of ambiguities.) Quote: “Our results indicate that under realistic conditions of measurement noise and model mismatch, estimating the orientation of a space object uniquely from its light curve is unlikely. Even if attitude inversion indicates only one solution, the non-convex nature of the problem makes it difficult to determine if one or more better solutions were missed.”

1.15 Encroachment analysis, claim by claim

1.16 The most dangerous overlap, in detail

The single passage in RF25 that comes closest to our framing is the description of the *axisymmetric* rocket body Case 1a (p. 23–25): they observe continuous 1D ω -bands and identify them as preserving the precession rate while varying the spin rate. The precession rate is $\omega_p = |\omega(0)|J_t/J_a$ (their Eq. 14). Notice: *this is a function of $|\mathbf{L}|$ at fixed inertia*. The spin rate ω_s varies along the band, and the body-frame polhode (for axisymmetric inertia) is a circle in the ω -plane perpendicular to the symmetry axis, traced at the precession rate while the body spins at ω_s about the symmetry axis. The cat-2 axisymmetric phenomenon they describe *is literally* “LC depends on $(|\mathbf{L}|, 2T)$ only, modulo spin phase about the symmetry axis.”

The danger: a careless reader (or referee) sees our cat-4 framing and says “isn’t this just RF25’s Case 1a generalised?” We need a clean answer to that critique, and the answer has to live in two places:

1. **For axisymmetric inertia**, body-frame polhode-invariance *plus visual rotational symmetry of the body* together produce the continuous spin-rate ambiguity RF25 observes (cat-2 + axisymmetric inertia). The body-frame-polhode geometry alone cannot explain it without the visual symmetry — because for asymmetric body geometry on the same axisymmetric inertia, the spin rotation would still change the LC by rotating non-symmetric facets through the body-frame sun/observer directions.
2. **For asymmetric inertia + asymmetric body geometry** (our m048 regime), there is no symmetry to invoke. The body-frame polhode is not a circle, the rotation has no axis of symmetry, and yet *cluster_457 on seed 89 has $|\mathbf{L}|$ matching truth to 0.55% but $\mathbf{L}_{J2000}$ -direction differing by 128.6°* . This is the empirical observation that RF25 has not made and that, if it converts to a class-level finding, fills the gap.

The key insight is that the *rotational degree of freedom in inertial space* (orienting $\mathbf{L}_{J2000}$ in $J2000$) is logically independent of any body symmetry. RF25 collapses it for the axisymmetric body because their particular case has visual symmetry; on an asymmetric body the same inertial-rotation DOF should still exist (because the LC depends on body-frame geometry only) but RF25 do not look for it. Our s073 measurement is consistent with that DOF being present on asymmetric bodies.

1.17 What RF25 does *not* provide

A precise inventory of the gaps in RF25 that the s073/cat-4 framing could fill:

- **No use of body-frame Casimir invariants.** “Polhode” does not appear anywhere in RF25. “Conservation” is mentioned only in the context of the BRDF being energy-conserving. The body-frame $(|\mathbf{L}|, 2T)$ structure is not invoked.
- **No computation or plotting of $\mathbf{L}_{J2000}$ direction across solutions.** This is the most striking absence. They have 10^5 converged solutions across four cases and never plot the inertial $\mathbf{L}_{J2000}$ direction. Our s073 plot (the one we should produce as a follow-up) is a direct fill of this gap.

- **No analytical proof of the body–frame–only LC dependence extended to L–direction ambiguity.** BRF24 §4 proves it for the discrete flip; the continuous version (claim A above) follows by the same argument applied to any inertial rotation that preserves $(\hat{u}, \hat{s})$, but RF25 / BRF24 do not write that argument down.
- **No multi–anchor / two–epoch architecture.** RF25 is purely single–epoch initial–condition optimisation. The s073 idea of using $\mathbf{L}_{J2000}$ conservation at two anchors as a discriminator (s060) is orthogonal to their approach.
- **No connection to the asteroid–LC literature** (Kaasalainen 2001), which is the only place where a continuous–L–family discussion exists in the literature. They cite the Frueh-group papers heavily but not Kaasalainen.

1.18 Verdict on encroachment

RF25 does not pre–empt the cat–4 framing. It documents adjacent phenomena (cat–2 continuous ω –bands on axisymmetric bodies) using machinery (BFGS clouds, 10^5 starts) that should have produced the cat–4 observation if it exists, but *they did not analyse their solutions through the body–frame Casimir lens that would surface it.*

The publication path remains open under two conditions, neither of which is secured yet:

1. **Empirical: cat–4 must be more than $N=1$.** The cohort sweep proposed in s073c step 3 (other Band–A multi–sols across seeds 10, 14, 28, 91, ...) must show the pattern repeats. If only seed 89’s cluster_457 exhibits the $(|\mathbf{L}|, 2T)$ –match with $\mathbf{L}_{J2000}$ mismatch, the framing is one anomaly.
2. **Theoretical: the analytical proof of claim A’s continuous version.** A few lines of algebra showing that for any rotation $R \in SO(3)$ acting on $\mathbf{L}_{J2000}$ that preserves the inertial sun and observer directions over the observing arc, the LC is invariant. The argument is that the LC depends on body–frame $(\hat{s}_{\text{body}}(t), \hat{o}_{\text{body}}(t))$, which is determined by the body–frame trajectory $q_{\text{body}}(t)$ around $\mathbf{L}_{J2000}$; rotating $\mathbf{L}_{J2000}$ in inertial space and simultaneously rotating the initial body orientation by the same R leaves the body–frame trajectory unchanged. This is the analogue of BRF24’s flip argument, written for arbitrary rotations of $\mathbf{L}_{J2000}$ rather than the single 180° rotation about the bisector. It should be a half–page derivation.

The risk if we ignore RF25. A referee will ask “how is this different from RF25 Case 1a?” The answer is: “RF25 Case 1a’s continuous ambiguity is cat–2 (visual + dynamical symmetry); ours is cat–4 (dynamical only); the discriminator is whether the body shape carries the symmetry, and our box–wing m048 geometry does not.” We need to make this distinction front and centre in any writeup.

1.19 Robinson & Frueh 2025 ECSD 2025 conference paper [RF#74]

RF25 cite their own ECSD 2025 paper ([RF74]: “Optimal light curve attitude inversion with measurement noise: two case studies”) as a companion work. That paper is not in our possession. *Action item:* check ECSD 2025 proceedings for the cited paper; it may add to or weaken the encroachment assessment, particularly if it explicitly treats asymmetric–inertia multi–solution structure.

Bottom line, both parts together

Part 1 (implementation). The single highest–value adoption is the NLL loss with σ_k weighting and signal rescaling (Eq. (??)). Cheap to implement, well–motivated, addresses the bright–peak overfitting we have seen empirically. Second–tier: the shadow cache (if forward–model wall

becomes the bottleneck post-elliprij), the α -overfit gate, the inertia ratio visualisation, and the body-frame ω scatter plot. Pass on: MRPs, Sutherland-Hodgman, BFGS swap.

Part 2 (encroachment). RF25 does not close the cat-4 framing. The adjacent finding they do publish — continuous ω -bands on axisymmetric bodies (Case 1a) — is cat-2 with axisymmetric inertia and a visually-symmetric body, not the cat-4 hypothesis. The publication path is still open conditional on (i) showing cat-4 holds beyond $N=1$ via the cohort sweep, and (ii) writing the half-page analytical extension of BRF24's flip argument to continuous rotations of $\mathbf{L}_{J2000}$. RF25's [RF74] companion paper is a residual risk and should be obtained.

Next compute, if Part 2 conclusions are accepted. Two cheap actions:

1. Plot $\mathbf{L}_{J2000}$ -direction for the converged clusters of seed 89 (s069 results), seed 10 (s069), seed 14 (s068), seed 28 (existing surveys) — all already in checkpoints. Re-scoring + plotting, no new BFGS, ≤ 30 min total. This is the cohort sweep step 3 from s073c, on cached data.
2. Write the analytical extension of BRF24 §4 to continuous $\mathbf{L}_{J2000}$ rotation (Part 2 condition ii). No compute, $\sim$ an hour of derivation, can be done in the same session.

If both clear, the framing has moved from $N=1$ anecdote to cohort-validated class-level claim plus theorem — and a writeup is in scope.

Table 1: Each s073 claim versus what RF25 actually states.

Claim	What RF25 actually states
A. <i>LC depends on body-frame geometry only</i>	Not Low. Use as setup, not finding. in novel their— ref- the er- body- encesframe- (BRF24) is de- cited pen- as dence the is source the of ac- the knowl- flip- edged problem start- the ing bodypoint frame of only the de- Frueh- pen- group dencepro- is gramme. the Any- ex- thing plicitwe foun-say da- about tion it of has BRF24). RF25cite do BRF24. not re- state it. RF25Part III is the gap our framing fills, if the demonopen. stratRF25 con- demon- tin- strate u- cat- ous 2 1- (ax- paramisym- bandsmet- in ric) body-con- framain- ω u- for ous the ω- ax- bands isym-but met- ex- ric plic- case itly
B. <i>Continuous 2-parameter $\mathbf{L}_{J2000}$-family on asymmetric bodies</i>	